Study title: Bridging resilience factors and dimensions of successful ageing: A cross-sectional network analysis

Assignment	Final Report for PL4246: Networks in Psychology
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Materials	The materials referenced in this report are available online through this link. This includes the Wave 2 National Social Life, and Ageing Project (NSHAP) dataset, the codebook, loaded objects, and knitted results.
Generative AI Declaration	This essay was reviewed and edited using Grammarly and OpenAI GPT-5.2 for spelling and grammar.
Presentation	A short presentation for this study is available through this link.

Abstract

Background: Resilience is often treated as a unitary construct, which may obscure differences in how specific resilience-related factors relate to the dimensions of successful ageing. **Methods**: Data from 3,196 older adults in the National Social Life, Health, and Aging Project (NSHAP) were used to estimate a network of resilience-related factors and successful ageing dimensions. Community structures were estimated and evaluated with modularity (Q). Bridge-Expected Influence (BEI) analyses assessed which resilience-related factors formed influential bridges with the successful ageing community. **Results and Conclusion**: The undirected, weighted cross-sectional network showed moderate connectivity (density = 0.39) and excellent edge stability (CS coefficient = 0.75). The walktrap and Louvain algorithms produced identical community partitions (Q =0.36) and identified four interpretable communities (physical functioning, cognitive and socioeconomic, relational health, mental well-being). Influential bridging nodes included mental health (BEI = 0.443), educational attainment (BEI = 0.436), and physical activity (BEI = 0.339).

Word count: 146

Bridging resilience factors and dimensions of successful ageing: A cross-sectional network analysis

Population ageing is a defining demographic shift of the 21st century. As life expectancy increases and fertility rates decline, a growing proportion of the population is entering older adulthood. Current projections indicate that by 2050, over 2.1 billion people will be aged 60 years or older (Lin et al., 2020). This demographic shift has substantial societal implications, including an increased demand for healthcare and social support (Kim & Park, 2017). This also creates the need for studies identifying modifiable pathways that can promote favourable health trajectories in later life. Older adults face a host of age-related challenges, some of which threaten well-being and physical functioning. For example, chronic disease burden, multimorbidity, functional decline and cognitive impairment may erode quality of life in older adults (Waddell et al., 2024). Furthermore, this life stage is often accompanied by notable transitions due to retirement, bereavement, and reduced social contact (Luo et al., 2012). However, these age-related challenges may not necessarily lead to declines in functioning and well-being. This suggests that age-related changes in functioning and well-being are not inevitable. Therefore, understanding how to promote favourable health trajectories in later life has become a public health priority.

Successful ageing (SA) is a gerontological framework for conceptualising favourable outcomes in older age. Introduced by Rowe and Kahn (1987), it is a multidimensional construct, encompassing an absence of chronic disease, an absence of difficulties activities of daily living (ADLs), high cognitive and physical function, and continued social participation. The concept of successful ageing has been influential, because it captures measurable age-related outcomes relevant to public health. For example, successful ageing has been linked to lower societal burdens through reduced healthcare spending, delayed retirement, and sustained productivity (Gianfredi et al., 2025). However, these outcomes vary considerably between individuals (Thoma et al., 2021), raising the question of why some individuals age successfully, while others experience substantial decline.

Resilience is defined as the capacity to maintain, recover, or adapt in response to adversity (Merchant et al., 2022). Differences in resilience capacity have been proposed as a mechanism that may explain heterogeneity in later life, and in health outcomes. In the context of ageing, sources of adversity include age-related internal challenges, such as multimorbidity or functional decline, and external challenges, such as social losses or environmental constraints. Under a resilience framework, successful ageing may be the result of multiple pathways that buffer the influence of age-related stress. Therefore, the ability to manage or adapt to these challenges may be the key to preserving functioning and well-being. This is consistent studies that have related successful ageing to resilience-promoting factors, such as physical activity (Vitor et al., 2023), and resilience-reducing factors, such as loneliness (Huisman, Klokgieters, & Beekman, 2017).

However, resilience is often operationalised as a unitary construct (Hu, Zhang, & Wang, 2015), which may conceal meaningful differences between specific resilience factors (Jacob et al., 2025). Individuals with high overall resilience capacity may have resources in some resilience factors (e.g. physical decline) and not others (e.g. social capital). If distinct resilience-related factors connect differentially to specific dimensions of successful ageing, collapsing them into a unitary construct may obscure these differences (Cosco, Howse, & Brayne, 2017), in ways that are potentially significant for intervention design. Therefore, examining resilience-related factors independently, and how they relate to each dimension of successful ageing, may be more informative (Jacob et al., 2025).

Network approaches are increasingly deployed in ageing research paradigms (Koivunen et al., 2024) and may be used to explore these interdependencies. Investigating cross-domain networks are also a powerful method for hypothesis-generation (Borsboom et al., 2021). Therefore, the present study had three aims. First, it aimed to examine how resilience factors formed a network with the dimensions of successful ageing. Second, it aimed to identify cross-domain communities within this network, using community detection algorithms. Third, it aimed to identify which resilience-related factors act as key bridging nodes with successful ageing in general.

Methods

Participants

This study was a cross-sectional, secondary analysis of data from Wave 2 of the National Social Life, Health, and Aging Project (NSHAP), a nationally-representative study of ageing in the United States (Suzman, 2009). It consists of 3,196 community-dwelling adults with sufficient data for the current research aims.

Variable selection

Successful ageing dimensions were derived from the Rowe-Kahn SA framework (Rowe & Kahn, 1987). First, chronic disease burden was measured as the total number of physician-diagnosed health conditions (out of 11 possible), following Vasilopoulos et al. (2014). Second, difficulties with activities of daily living (ADLs) were measured in NSHAP (Huisingh-Scheetz et al., 2014) as the number of difficulties reported across 11 ADLs. In anticipation of bridge analyses, both chronic disease burden and difficulties with ADLs were reverse-coded, so that higher scores indicated fewer health conditions or fewer difficulties in ADLs. Third, physical health in NSHAP (Cullati et al., 2020) was self-rated on a 5-point scale (1 = “poor” to 5 = “excellent”). Fourth, cognitive function was evaluated using the Montreal Cognitive Assessment (MoCA), a measure of global cognition in older adults. And finally, social participation in NSHAP (Waite, Duvoisin, & Kotwal, 2021) was assessed based on the frequency of socialising with friends or relatives in the past year, rated on a 7-point scale (0 = “never” to 6 = “several times a week”).

Resilience-related factors were identified based on existing research highlighting both resilience-promoting and resilience-reducing factors in later life (Brinkhof et al., 2025; Koivunen et al., 2024). Five measures of mental health: mental health, depression, anxiety, happiness, and loneliness, were included as resilience factors. These mental health measures are described in detail by Payne et al. (2014). Two measures of social health, community participation and the number of friends, were also included (Waite, Duvoisin, & Kotwal, 2021). Educational attainment was also included, as an indicator of cognitive reserve (Suchy-Dicey et al., 2025). Four lifestyle factors, including smoking, alcohol consumption (Drum et al., 2009), physical activity, and body-mass index (Cho, 2024) were included as resilience factors. Two demographic variables, age and income, were included. No variables displayed variable redundancy with the goldbricker() function from the network tools package (Jones, 2017). This indicates that there was no significant conceptual overlap among the variables. While some variables were derived from item-level ordinal measurements, all items had at least 5 response categories. This satisfies the recommended criteria for modelling them as approximately continuous variables (Rhemtulla, Brosseau-Liard, & Savalei, 2012).

Preprocessing and missing data

All analyses were conducted in *RStudio* (Version 4.5.1). Missing values, which comprised about 4.4% of the dataset, were imputed via random forest imputation using the missRanger package (Mayer & Mayer, 2019). Skip-listed items were recoded prior to imputation, as questionnaire routing in NSHAP may induce variable dependencies if imputed (Burger et al., 2023). The codes required to distinguish genuinely missing or skip-listed responses were sufficiently described in the codebook.

Network estimation procedure

An undirected, weighted network was estimated using *bootnet* (Epskamp, Borsboom, & Fried, 2018). In this network, the nodes correspond to the variables included, and edges correspond to partial Spearman correlations that represent conditional associations between each pair of nodes while controlling for all others. Given variable skewness, detected with the *dlookr* package (Staniak & Biecek, 2019), edges were estimated as partial Spearman correlations. Given the large sample size (N = 3,196) relative to the number of nodes (N = 19), assumptions of network sparsity may not hold. This is as penalised regularisation, which prioritises sparsity, may risk omitting meaningful edges (Isvoranu & Epskamp, 2023). Therefore, a model selection procedure was applied using *ggmModSelect()* to prioritise edge specificity (Isvoranu & Epskamp, 2023). The network was visualised using *qgraph* (Epskamp et al., 2012). The Fruchterman-Reingold algorithm was used to generate a layout where nodes with stronger and more numerous connections were placed more centrally. Node-wise predictability was estimated using *mgm* (Haslbeck & Waldorp, 2020) and represents the proportion of variance in each node explained by its neighbours in the network (R^2). Given that coherent subsystems linking resilience factors and successful ageing dimensions are likely (Merchant et al., 2022; Trica et al., 2024), community detection was performed. Community structures were identified using *walktrap*, *Louvain*, and *fastgreedy* algorithms within *igraph* (Csárdi et al., 2006), with partitions evaluated with modularity (Q).

Bridge centrality

Bridge expected influence (BEI) was computed in *networktools* (Jones, 2017) and represented the signed sum of a node's edge weights connected to nodes outside its community. Consistent with the aims of this study, BEI values of resilience-related factors were examined to influential bridges that connect most strongly to successful ageing dimensions. Nodes with BEI values above the 80^{th} percentile were identified as influential bridges, as recommended by Jones, Ma and McNally (2021). BEI values were examined as 1-step BEI, which summarises immediate cross-community edges only. Additionally, as BEI is a signed metric, and is read as the net balance of positive and negative cross-domain conditional associations, how it is interpreted depends on how node values are computed (Jones, Ma, & McNally, 2021). For this reason, all successful ageing dimensions were coded non-arbitrarily, such that higher values in each dimension were associated with more successful ageing.

Stability and accuracy

Edge weight stability was assessed via nonparametric bootstrapping with 1,000 bootstrap samples in bootnet (Epskamp, Borsboom, & Fried, 2018). Bridge expected influence (BEI) stability was estimated using case-dropping bootstraps and summarised with the correlation stability (CS) coefficient. Bootstrapped difference tests were conducted to compare edge weights and centrality measures (e.g. BEI) between nodes.

Results

Network description

The estimated undirected, weighted network is displayed in Figure 1. The network comprised 19 nodes and 67 non-zero edges out of 171 possible edges, indicating moderate connectivity and a density of 39.2%. The correlation stability coefficient of edge weights was excellent (CS = 0.75), indicating that the estimated network structure was robust to sampling variation under the bootstrap procedure

(Epskamp, Borsboom, & Fried, 2018). The average node-wise predictability of the cross-domain network was $R^2 = 0.26$. The most predictable nodes were physical health ($R^2 = 0.43$), depression ($R^2 = 0.41$), and cognition ($R^2 = 0.39$). The least predictable nodes were smoking ($R^2 = 0.05$), alcohol use ($R^2 = 0.09$), and body mass index ($R^2 = 0.11$).

Community detection

For community detection, that walktrap and Louvain yielded identical partitions, and had the highest modularity ($Q = 0.36$) (Csárdi et al., 2006). The communities were interpretable (see Figure 1): the physical and functional capacity community comprised 7 nodes and included the absence of chronic diseases (SA), the absence of difficulties in ADLs (SA), physical health (SA), physical activity, smoking, age, and body-mass index. The cognitive and socioeconomic community comprised 4 nodes and included cognitive functioning (SA), alcohol consumption, educational attainment, and household income. The relational health community comprised 3 nodes and included social participation (SA), number of friends, and community participation. The mental well-being community comprised 5 nodes and included happiness, mental health, depression, anxiety, and loneliness. All communities, except the mental well-being community, were cross-domain, and had nodes from both the successful ageing and resilience communities.

Bridge centrality analyses

Bridge expected influence (BEI) estimates had excellent stability (CS = 0.75). BEI analyses were conducted between the resilience factors and the predefined Rowe-Kahn successful ageing community. Resilience-related factors that formed influential bridges with successful ageing were mental health (BEI = 0.443), educational attainment (BEI = 0.436) and physical activity (BEI = 0.339). Bootstrapped difference tests also indicated that these nodes formed significantly stronger bridges than the next-ranked nodes.

Discussion

This study examined how resilience-promoting and resilience-reducing factors contribute to successful ageing through distinct pathways. The findings suggest that resilience-related factors may not act as a single unified construct when it comes to successful ageing. Instead, resilience factors may form multiple coherent subsystems, spanning physical function, cognition and socioeconomic resources, relational health, and emotional well-being. This pattern supports the view that resilience may be better conceptualised as a network of specific protective and risk factors than as a single composite index (Merchant et al., 2022).

The identified community structures suggest that resilience-related factors may cluster with specific successful ageing dimensions. This cross-domain clustering may reflect discrete late-life challenges such as specific disease syndromes, social isolation, or vice behaviours (Jacob et al., 2022), which organise into particular stressor(s), adaptive resources, and relevant outcomes. This clustering also aligns with the notion that there are multiple successful ageing phenotypes (Rowe & Kahn, 2015). From a systems perspective, this suggests that targeted interventions for specific successful ageing outcomes may target resilience factors that it forms a community with.

Bridge expected influence (BEI) analyses identified mental health, education, and physical activity as the most influential bridges between resilience factors and successful ageing outcomes. By relating more broadly to successful ageing, these may constitute potential intervention targets in the general promotion of successful ageing. Mental health maintains a bidirectional relationship with successful ageing, by influencing motivation, self-care, and healthcare engagement (Ohrnberger, Fichera, & Sutton, 2017). Educational attainment represents cognitive reserve and lifelong cognitive enrichment (Suchy-Dicey et al., 2025), which is known to buffer the effects of age-related neurobiological changes on cognition (Stern et al., 2023). Physical activity is consistently associated with functional performance, disability prevention, and aspects of cognitive and functional performance (Stuart, 2023; Tang et al., 2025; Vitor et al., 2023). It may serve as a behavioural pathway through which social and psychological resources translate into preserved functioning (Weening-Dijksterhuis et al., 2011). Together, these bridging factors may be promising candidates for multi-domain interventions designed to support the general promotion of successful ageing.

However, these findings should be considered in light of several limitations. First, the estimated network is cross-sectional which precludes causal inference (Pearl, 2014). For example, it is plausible that physical activity is downstream of worsening cognitive function, chronic disease burden, and difficulties with ADLs (Lin et al., 2020). Future longitudinal approaches, such as cross-lagged panel network models (Chen et al., 2025), are needed to confirm whether these resilience factors prospectively predict changes in the dimensions of successful ageing. Second, a common critique of network analysis is that conditional associations revealed within a network may not necessarily mean the relationship exists in the population (Neal et al., 2022). For example, these conditional associations are highly sensitive to the set of variables included. This study attempted to include a comprehensive selection of known resilience factors based on the limited network studies available (Brinkhof et al., 2025; Chen et al., 2025). However, it remains plausible that other resilience factors that were not available in the National Social Life, Health, and Aging Project (NSHAP), such as coping mechanisms (Lazarus, 1966), and internalised age stereotyping (Brinkhof et al., 2025), may change the results of this study. Third, the findings of these bridge analyses should be used to prioritise hypotheses for future longitudinal or interventional testing, rather than evidence of causation. Lastly, the bridge analyses were examined between resilience factors and a predefined community of successful ageing conceptualised by Rowe and Kahn (1987). Currently, there is no widely accepted conceptualisation of successful ageing (Cosco, 2015; Huisman, Klokgieters, & Beekman, 2017; Waddell et al., 2024). Therefore, these cross-domain findings may differ based on how successful ageing is defined.

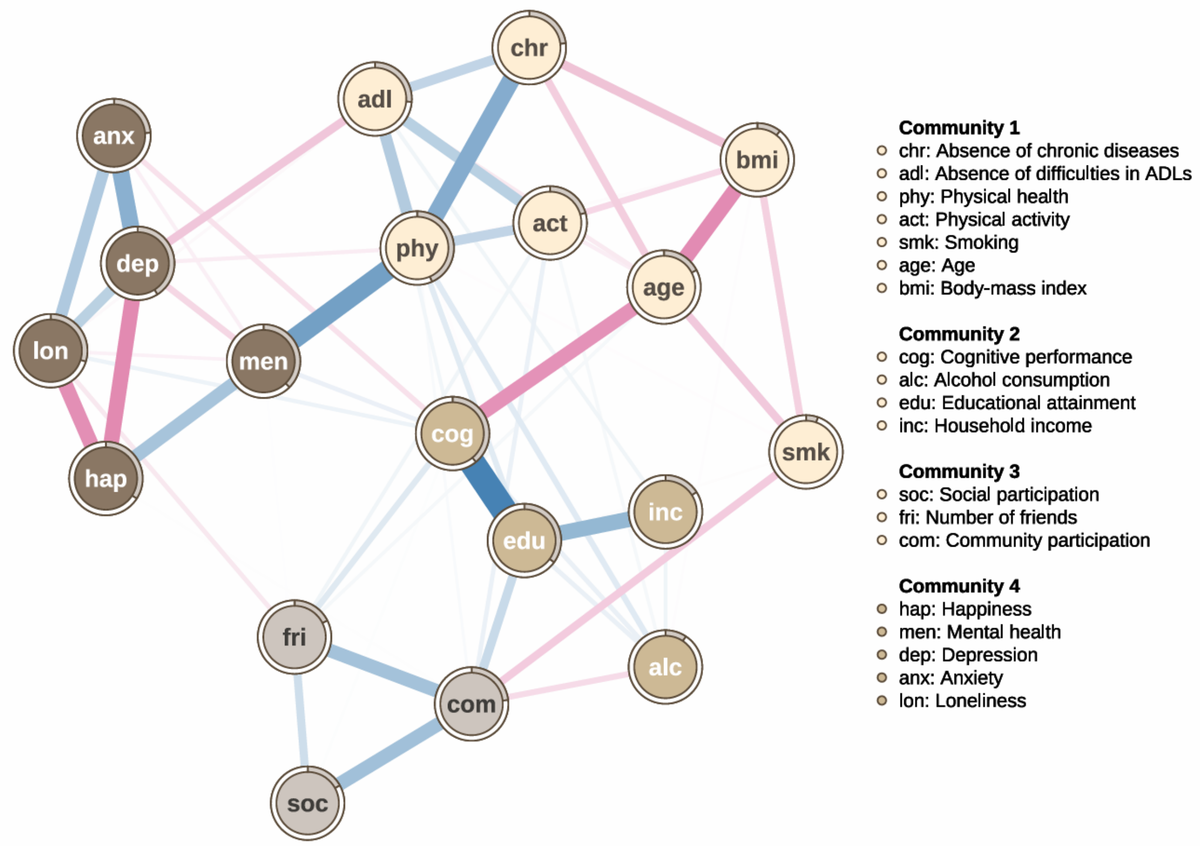
Future research should examine whether the bridging nodes identified replicate in other cohorts, and under alternative definitions of successful ageing. Longitudinal network approaches, such as cross-lagged panel networks (Chen et al., 2025), may be used to clarify the causal direction of these patterns of connectivity. Intervention studies may test whether bridging nodes yield broad improvements in the general promotion of successful ageing, and whether targeted interventions on specific resilience factors may be used to target successful ageing outcomes within its community. Additional work could also examine subgroup differences, such as sex, younger age groups, and within the context of disease and disability (Jones, Ma, & McNally, 2021).

Conclusion

This cross-sectional network analysis suggests that resilience-related factors and successful ageing dimensions form an interconnected system with multiple communities. These communities may represent distinct pathways to preserving late-life functioning and well-being, and supports the view that resilience may not be reducible to a unitary construct (Hu, Zhang, & Wang, 2015), especially when investigating successful ageing outcomes. This study identified mental health, educational attainment, and physical activity as influential cross-domain nodes that bridge resilience factors and successful ageing dimensions. These may be used as plausible intervention targets for future longitudinal research, for designing multidomain interventions aimed at promoting successful ageing.

Figure 1

Network Plot of Successful Ageing Dimensions and Resilience Factors



Note. The network plot visualises partial Spearman correlations between variables. Nodes represent each variable, with grey pies around each node representing the proportion of variance explained (R^2) by all other nodes in the network. Edges represent pairwise partial correlations after controlling for all other nodes. Edge colour indicates the sign of the association (blue = positive, red = negative), and edge width is proportional to the magnitude of the association.

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